

Wind Turbine Gearbox Vibration Monitoring and Anomaly Detection Using Data Analytics

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Abstract— This study presents a vibration-based anomaly detection system for wind turbine gearbox condition monitoring using engineering data analytics and machine learning techniques. The system was developed to improve predictive maintenance capability by detecting abnormal gearbox behavior at an early stage. Vibration telemetry data from multiple sensor axes together with gearbox bearing temperature measurements were collected and processed through a modular Python-based analytics pipeline. The workflow integrates data preprocessing, statistical analysis, anomaly detection, and visualization into a unified monitoring framework. Statistical metrics including mean, median, variance, standard deviation, and skewness were utilized to evaluate gearbox operational behavior and vibration characteristics. An unsupervised Isolation Forest algorithm was implemented to identify abnormal operating conditions associated with excessive vibration, thermal irregularities, and possible mechanical degradation. Visualization techniques such as time-series analysis, correlation heatmaps, boxplots, and interactive dashboards were used to support engineering interpretation and predictive maintenance analysis. Results indicate that the Z-axis vibration exhibited the highest operational variability and strongest indication of instability. The anomaly detection model successfully distinguished normal and abnormal operating states, demonstrating the effectiveness of combining vibration monitoring with machine learning analytics. The proposed system improves monitoring automation, enhances gearbox reliability, reduces maintenance costs, and supports scalable condition-based maintenance applications for renewable energy systems.

Keywords— Wind turbine gearbox, vibration monitoring, anomaly detection, predictive maintenance, machine learning, Isolation Forest, engineering data analytics

I. INTRODUCTION

Renewable Energy is important because it provides clean electricity without producing carbon dioxide during operation, helping reduce greenhouse gas emissions and slow climate change. Wind Energy also improves energy security by reducing dependence on fossil fuels while creating sustainable and affordable power for growing global energy needs [1]. Gearbox maintenance in tidal current energy converters (TCECs) is particularly challenging and costly because gearboxes operate under extreme underwater conditions, including high loads,

corrosion, fluctuating tidal forces, and limited accessibility for inspection and repair. In most cases, maintenance activities require specialized vessels, divers, or heavy lifting equipment to retrieve submerged components, leading to long downtime periods and significantly increased operation and maintenance (O&M) costs [2].

Gearbox failures are difficult to detect early because operating conditions produce complex and noisy SCADA and CMS data, while limited failure examples and gradual internal degradation reduce the accuracy and robustness of predictive models [3]. Intelligent monitoring systems are essential in wind turbines because traditional manual monitoring methods are slow, labor-intensive, and unable to provide real-time fault detection. By combining advanced sensors with artificial intelligence and machine learning techniques, intelligent monitoring improves early fault detection, enhances reliability, and reduces maintenance costs and unexpected downtime [4].

The main objective of this study is to develop a vibration-based anomaly detection system for gearbox condition monitoring using data analytics techniques. Specifically, the study aims to collect vibration signals from gearbox operation, extract meaningful vibration features, apply data analytics or machine learning techniques for data analysis, and detect abnormal gearbox behavior at an early stage to improve system reliability, reduce maintenance costs, and prevent unexpected failures.

II. LITERATURE REVIEW

Wind turbines operate under harsh and varying environmental conditions, making their components prone to mechanical stress and failure. Among these components, the gearbox is one of the most critical parts because it transfers rotational energy from the turbine blades to the generator. Continuous exposure to heavy loads, vibration, and fluctuating wind speeds can lead to faults such as gear wear, bearing damage, misalignment, and lubrication problems. To reduce maintenance costs and prevent unexpected failures, researchers have developed condition monitoring techniques such as vibration analysis and machine learning-based fault detection for early diagnosis and predictive maintenance [5].

A. Wind Turbine Gearbox System

A gearbox is an essential mechanical component that transmits power and adjusts torque and rotational speed between connected shafts through a system of gears, bearings, and shafts. Gears are used to transfer power efficiently, while bearings support rotating parts and reduce friction during operation. In wind turbines, the gearbox plays a critical role by increasing the low rotational speed of the turbine blades into the high speed required by the generator to produce electricity. Because wind turbines operate under harsh environmental conditions and continuous mechanical loading, gearbox components are highly exposed to wear, vibration, and stress, making condition monitoring and fault detection important for maintaining reliable turbine performance [6].

B. Vibration Monitoring Techniques

Accelerometers are widely used vibration sensors in rotating machinery because they can accurately measure linear acceleration and detect abnormal machine behavior. Proper sensor mounting is essential for accurate vibration data acquisition, with common mounting methods including stud, adhesive, and magnetic mounting depending on the application and monitoring requirements [7]. Time-domain analysis is a commonly used vibration monitoring technique that evaluates vibration signals over time using statistical parameters such as RMS, peak value, crest factor, and kurtosis to detect abnormalities and faults in rotating machinery such as imbalance, misalignment, bearing defects, and gear damage [8].

C. Common Gearbox Faults

Gearbox failures are among the most critical problems in wind turbine drivetrains because they result in significant downtime and economic losses. Common gearbox faults include gear tooth damage, bearing failures, shaft-gearbox decoupling, pitting, cracking, scratching, and fretting wear caused by stiffness disparities, fit resistance, and gear hub thickness effects. Bearings inside the planetary gearbox, particularly planet bearings, intermediate shaft-locating bearings, and high-speed locating bearings, are especially vulnerable to failure. In addition, poor lubrication is a major contributor to gearbox deterioration, with changes in oil viscosity, water content, particle count, and debris often serving as indicators of developing faults [9].

Failures in the pitch control system can also indirectly contribute to gearbox damage by creating operational instability and excessive drivetrain loads. Problems such as hydraulic leakage, early brake activation, reduced hydraulic fluid effectiveness, and asymmetry in pitch angle can decrease system stability and resilience, leading to abnormal operating conditions that accelerate gearbox wear and increase the risk of failure [5].

D. Data Analytics and Machine Learning in Fault Detection

Data analytics and machine learning improve fault detection by analyzing operational data to identify abnormalities and predict failures in real time. These technologies process large volumes of operational

data to detect abnormal patterns and potential system failures before they occur [10]. They also enhance the accuracy, speed, and reliability of condition monitoring in electrical power and control systems, leading to more efficient maintenance and reduced downtime [11].

III. SYSTEM ARCHITECTURE & MATHEMATICAL FRAMEWORK

A. System Architecture Design

This study utilized a modular Python-based Engineering Data Analytics System designed to automate the processing, analysis, and visualization of gearbox monitoring telemetry data through a unified sequential data pipeline architecture. The system integrates data ingestion, automated preprocessing, statistical analytics, anomaly detection, and visualization deployment into a single workflow to improve maintainability, scalability, and automation efficiency. It was developed using several scientific computing libraries, including Pandas for dataset manipulation and data cleaning, NumPy for numerical and statistical computation, Matplotlib and Seaborn for static visualization, Plotly for interactive and animated analytics dashboards, and Scikit-learn for machine learning-based anomaly detection using the Isolation Forest algorithm.

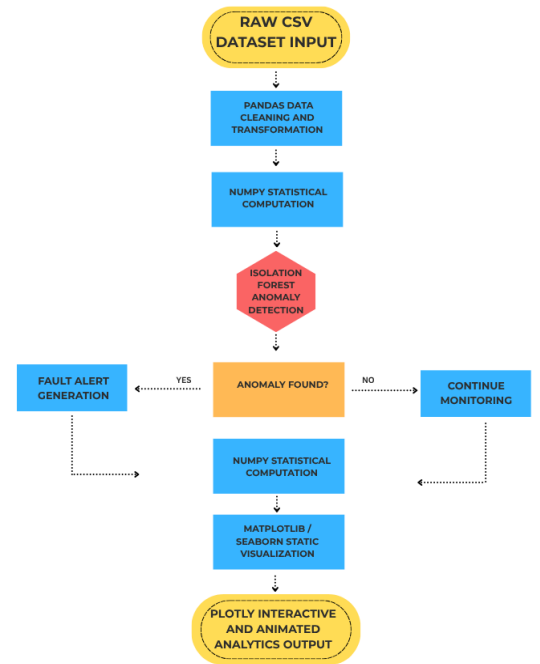


Fig. 1. Data Analytics and Machine Learning-Based Fault Detection System Flowchart

This workflow explains how wind turbine gearbox vibration data is analyzed to detect faults early. Raw CSV vibration data is cleaned and transformed using Pandas, then NumPy performs statistical analysis on the signals. An Isolation Forest model detects abnormal vibration patterns that may indicate gearbox faults. If an anomaly is found, a fault alert is generated; otherwise, monitoring continues. Finally,

Matplotlib, Seaborn, and Plotly create static and interactive visualizations to help engineers monitor gearbox health and support predictive maintenance.

B. Mathematical Definitions and Evaluation Strategy

The Engineering Data Analytics System utilized several statistical tools to analyze and summarize gearbox telemetry data. These methods were used to evaluate operational behavior, measure variability, and identify data distribution patterns within the monitored parameters. The formulas for the statistical metrics used in the study are presented below.

The mean represents the average value of a dataset and is used to determine the central tendency of gearbox telemetry measurements.

$$\bar{X} = \frac{\sum X}{n}$$

The median is the middle value of an ordered dataset and is useful for identifying the central location of data while reducing the effect of outliers.

$$x_{\frac{n+1}{2}} \quad \text{For odd } n: \quad x_{\frac{n}{2}} + x_{\frac{n}{2}+1} \quad \text{For even } n:$$

For odd n:

For even n:

Standard deviation measures the spread or dispersion of telemetry values from the mean. A higher standard deviation indicates greater operational variability.

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Variance quantifies the squared deviation of data points from the mean and is used to evaluate signal stability.

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

The minimum identifies the smallest recorded telemetry value. This metric helps determine the lower operational boundary of the system.

$$\min(x_1, x_2, \dots, x_n)$$

The maximum identifies the largest recorded telemetry value. This metric helps determine the peak operational condition observed in the gearbox system.

$$\max(x_1, x_2, \dots, x_n)$$

Skewness measures the asymmetry of the data distribution. Positive skewness indicates a longer right tail. Negative skewness indicates a longer left tail. Near-zero skewness indicates a nearly symmetric distribution.

$$\frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\sigma^3}$$

IV. DATA PROCESSING PIPELINE

A. Dataset Preparation and Characterization

The dataset utilized in this study consists of gearbox telemetry measurements collected from vibration monitoring sensors and thermal monitoring systems. The primary variables analyzed include vibration_x, vibration_y, vibration_z, gearbox_bearing_temp, and timestamp. These parameters were selected because they are critical indicators of rotating machinery health and operational stability.

The vibration measurements represent multi-axis sensor readings used to identify abnormal oscillations, imbalance, bearing degradation, or shaft misalignment. Meanwhile, the gearbox_bearing_temp variable provides thermal condition monitoring to evaluate overheating behavior associated with mechanical stress or lubrication failure.

TABLE I. SAMPLE GEARBOX TELEMETRY DATA

TIMESTAMP	VIBRATION X	VIBRATION Y	VIBRATION Z	GEARBOX BEARING TEMPERATURE
2025-01-01 00:00:00	0.0109	0.0092	.013	65.85
2025-01-01 00:00:01	0.0076	0.0065	0.0092	60.83
2025-01-01 00:00:02	0.016	0.0136	0.0193	65.16
2025-01-01 00:00:03	0.0164	0.014	0.0197	65.8
2025-01-01 00:00:04	0.0177	0.0151	0.0213	66.84

To preserve numerical consistency and analytical reliability, the dataset underwent rigorous preprocessing procedures. The system automatically converted timestamp fields into datetime format to enable temporal analysis and trend visualization. Duplicate observations were removed programmatically to eliminate redundant telemetry fragments.

Missing observations were handled using forward-fill interpolation techniques to preserve temporal continuity within the monitoring sequence. Numerical conversion routines were also implemented to coerce invalid string artifacts into NaN values, followed by automatic removal of incomplete rows to maintain computational integrity.

The cleaned dataset was subsequently exported into a secondary CSV file to ensure reproducibility and traceability of all preprocessing operations.

B. Computational Analytics Procedure

The Engineering Data Analytics System implemented multiple computational analytics modules to evaluate gearbox operational behavior and detect abnormal conditions. Robust exception-handling mechanisms were embedded throughout the pipeline to prevent runtime interruptions caused by corrupted telemetry values, invalid formatting, or missing files.

The statistical analytics module extracted all numerical features from the dataset and computed aggregate engineering metrics including mean, median, standard deviation, variance, minimum value, maximum value, and skewness. These metrics were

utilized to characterize vibration distributions and evaluate operational variability within the gearbox system.

Correlation analysis was subsequently performed using Pearson correlation coefficients to identify relationships between vibration amplitudes and thermal behavior. The generated correlation matrix was visualized through a heatmap representation to support engineering interpretation of inter-variable dependencies.

A comparative analytics routine was also implemented to evaluate vibration behavior under different thermal operating conditions. The system separated low-temperature and high-temperature operating states using a threshold-based classification approach and computed the mean vibration amplitude for each condition. This enabled assessment of the effect of temperature on gearbox dynamic stability.

C.. Anomaly Detection and Machine Learning Analysis

To enhance predictive maintenance capability, the system implemented an unsupervised machine learning approach using the Isolation Forest algorithm from the Scikit-learn library. The anomaly detection module analyzed multidimensional numerical telemetry features, including vibration amplitudes and gearbox bearing temperature, to identify abnormal operating conditions automatically.

The Isolation Forest model was configured with a contamination factor of 0.03, indicating that approximately 3% of the observations were expected to represent abnormal behavior. The algorithm isolates unusual telemetry patterns by recursively partitioning the dataset, allowing the system to distinguish normal operational behavior from potentially hazardous mechanical conditions.

After model execution, each telemetry record was assigned an anomaly classification label. Observations predicted as normal were labeled "Normal," while suspicious operating conditions were labeled "Anomaly." The resulting classifications were appended directly into the processed dataset to support further engineering diagnostics and visualization analysis.

This machine learning implementation enables rapid identification of unstable gearbox conditions associated with excessive vibration, thermal irregularities, bearing wear, shaft imbalance, or rotating component failure.

D.. Visualization and Monitoring Dashboard Generation

The visualization subsystem was developed to transform raw telemetry measurements into interpretable engineering graphics and interactive monitoring dashboards. Both static and dynamic visualization techniques were utilized to support engineering analysis and operational monitoring.

Static visualizations were generated using Matplotlib and Seaborn libraries. The system produced time-series graphs displaying vibration

trends across the X, Y, and Z axes over time. Correlation heatmaps were also generated to illustrate the relationships among numerical variables within the gearbox system. Additionally, boxplot visualizations were created to evaluate vibration distribution, spread, and the presence of outlier behavior.

To improve monitoring capability and user interaction, the system incorporated Plotly-based interactive dashboards. The interactive gearbox monitoring graph included range sliders, zooming functionality, and temporal filtering tools to enable detailed inspection of vibration behavior over different operating periods.

An animated scatter visualization was also implemented to display the relationship between gearbox bearing temperature and vibration amplitude dynamically over time. The animation incorporated anomaly labels generated by the Isolation Forest algorithm, enabling visual tracking of abnormal operational states throughout the telemetry sequence.

All visualization outputs were automatically displayed during execution and exported into the outputs directory for engineering documentation and further review.

E. Output Storage and File Generation

To ensure reproducibility and organized storage, the system automatically generated and exported multiple files throughout the execution pipeline. A dedicated output directory was created at runtime to store processed datasets, reports, and visualization results. The cleaned dataset and statistical summary tables were exported as CSV files for future analysis and documentation.

Static visualizations, including time-series graphs, correlation heatmaps, and boxplots, were automatically saved as PNG image files. Interactive Plotly dashboards were also exported as standalone HTML files, allowing users to access the monitoring system through a web browser without additional software requirements.

This automated file-generation process improves workflow efficiency, supports analytical reproducibility, and ensures that all engineering outputs remain accessible for reporting, maintenance evaluation, and future system validation.

V. RESULTS

A. Statistical Summary Tables

TABLE II. TEMPERATURE MEASUREMENTS STATISTICAL SUMMARY

FEATURE	MEAN	MEDIAN	STANDARD DEVIATION	MINIMUM	MAXIMUM	SKENNESS
gearbox_oil_temperature	61.32	61.25	7.48	45.68	87.59	0.10
gearbox_bearing_temperature	69.85	70.79	8.34	51.09	95.83	-0.03

The gearbox oil temperature has a mean of 61.32 and a standard deviation of 7.48, while the gearbox bearing temperature has a higher mean of 69.85 with a standard deviation of 8.34. The minimum and

maximum values show a moderate operating range for both features. The mean and median values are very close in both cases, and the skewness values are near zero.

TABLE III. VIBRATION MEASUREMENTS

FEATURE	MEAN	MEDIAN	STANDARD DEVIATION	MINIMUM	MAXIMUM	SKEWNESS
vibration_x	0.0198	0.0188	0.0059	0.0015	0.0479	0.8714
vibration_y	0.0169	0.0159	0.0050	0.0013	0.0407	0.8714
vibration_z	0.0238	0.0225	0.0071	0.0018	0.0574	0.8714

The vibration_x, vibration_y, and vibration_z features all have low mean values and small standard deviations, showing limited variation in vibration intensity. The z-axis records the highest mean and maximum vibration values among the three directions. All vibration features have the same positive skewness value of 0.8714.

TABLE IV. LUBRICATION AND OIL SYSTEM MEASUREMENTS

FEATURE	MEAN	MEDIAN	STANDARD DEVIATION	MINIMUM	MAXIMUM	SKEWNESS
oil_pressure	4.3454	4.35	0.1120	3.44	4.65	1.8313
particle_count	765.27	766.00	380.03	100.00	1438.00	0.0032
vibration_z	0.0238	0.0225	0.0071	0.0018	0.0574	0.8714

Oil pressure has a mean of 4.3454 with a low standard deviation of 0.1120, indicating stable pressure values, although the skewness is highly positive (1.8313). Particle count has a very large range and high standard deviation, reflecting significant variability in contamination levels. Vibration_z again shows moderate variability and positive skewness.

B. Static Graph Output Analysis

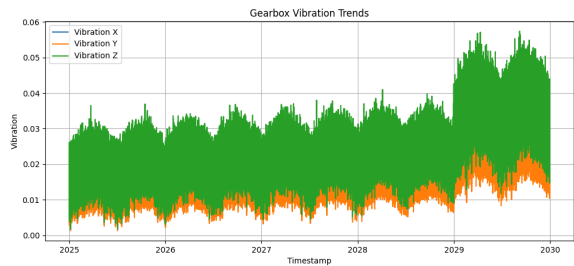


Fig. 2. Gearbox Vibration Trend Analysis Over Time

The Gearbox Vibration Trends graph shows a gradual increase in vibration levels from 2025 to 2030 across the X, Y, and Z axes. Among the three axes, the Z-axis consistently exhibits the highest vibration amplitude, followed by the X and Y axes. Around 2029, a noticeable rise in vibration magnitude occurs, especially in the Z-axis, where the values increase significantly compared to earlier years. The graph also displays recurring fluctuations over time, indicating periodic operational behavior or varying load conditions. Additionally, the vibration signals become more dispersed toward the later years, reflecting increasing instability in the gearbox system.

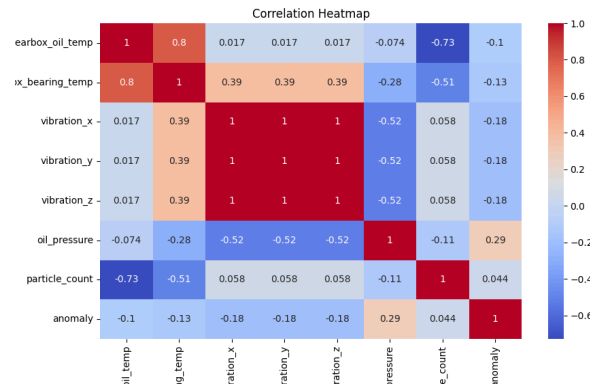


Fig. 3. Correlation Heatmap of Gearbox Parameters

The Correlation Heatmap illustrates the relationships between gearbox variables. Gearbox oil temperature and bearing temperature show a strong positive correlation of approximately 0.80, indicating that both variables increase together. The vibration variables (X, Y, and Z) exhibit nearly perfect positive correlations with one another, suggesting highly similar vibration behavior across all axes. Oil pressure has a moderate negative correlation with the vibration variables (-0.52), meaning that vibration tends to increase as oil pressure decreases. Bearing temperature also has a moderate positive relationship with vibration (0.39), while anomaly detection shows only weak correlations with most parameters.

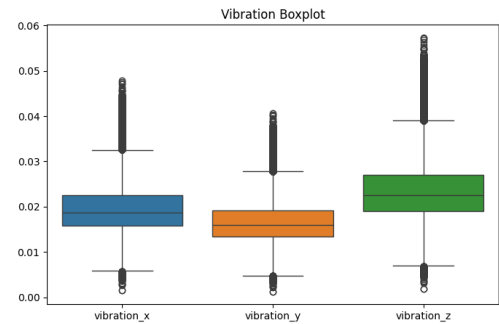


Fig. 4. Boxplot of Gearbox Vibration Distribution

The Vibration Boxplot presents the distribution of vibration values for the X, Y, and Z axes. The Z-axis has the highest median vibration value and the widest spread among the three axes, indicating greater variability in vibration behavior. Numerous outliers are present in all axes, especially in the Z-axis, where extreme vibration values are more frequent. The distributions are positively skewed, meaning that most vibration values remain within a moderate range while occasional high spikes occur. The larger interquartile range in the Z-axis further indicates greater instability compared to the X and Y axes.

C. Animated Graph Output Analysis

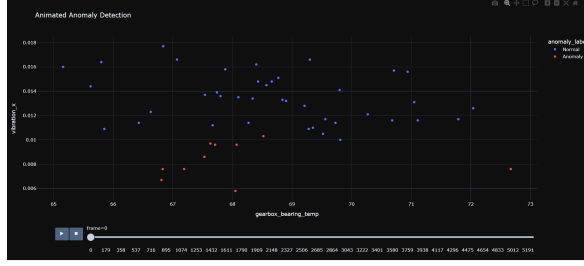


Fig. 5. Animated Scatter Graph of Anomaly Detection

The animated scatter plot presents the relationship between gearbox bearing temperature and vibration in the X-axis direction. Normal operating conditions are represented by blue markers, while anomalous conditions are shown in red. Most normal data points are distributed within a vibration range of approximately 0.010 to 0.018 and a temperature range between 65°C and 72°C. In contrast, anomaly points appear concentrated at lower vibration values, mainly between 0.005 and 0.010, and are clustered around temperatures of 66°C to 68°C, with a few isolated anomalies at higher temperatures. The animation component allows the monitoring of changes over time, enabling users to observe how operational conditions evolve dynamically. This temporal representation improves understanding of how anomalies emerge and whether they occur suddenly or progressively during turbine operation.

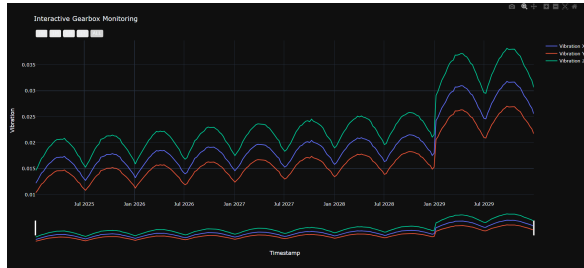


Fig. 6. Interactive Gearbox Monitoring Time-Series Plot

The interactive time-series graph illustrates the variation of vibration amplitudes along the X, Y, and Z axes over an extended operational period from 2025 to 2029. The vibration signals display periodic oscillations, indicating cyclical operational behavior influenced by rotating gearbox components and changing turbine loads. Among the three axes, the Z-axis vibration consistently exhibits the highest amplitude, followed by the X-axis and Y-axis vibrations.

A gradual upward trend is observed in all vibration components over time, with a significant increase occurring around early 2029. Following this point, vibration amplitudes rise sharply, reaching peak values close to 0.038 in the Z-axis direction. The interactive controls and range slider further enhance data exploration by allowing users to zoom into specific operational periods for detailed analysis.

VI. TECHNICAL DISCUSSION

A. Description of the Statistical Summary Tables

Table 1 results indicate that both gearbox oil and bearing temperatures are stable and consistently distributed during operation. The near-zero skewness suggests the absence of major abnormal temperature fluctuations, implying normal and balanced thermal conditions within the gearbox system.

Table 2 has vibration signals that are generally stable, but the positive skewness indicates occasional spikes in vibration levels. Since vibration_z has the highest readings, the machine may experience stronger movement or mechanical stress along the z-axis, which could require monitoring for possible imbalance or wear.

Table 3 results of the oil pressure system are mostly stable but occasionally experience unusually high readings. The high variability in particle count may indicate inconsistent contamination or wear debris within the system, which should be monitored closely. The repeated positive skewness in vibration_z confirms intermittent increases in vibration that may signal developing mechanical issues.

B. Description of the Static Graph Output

Figure 2 shows the increasing vibration trend which suggests progressive mechanical degradation or wear within the gearbox components over time. The dominance of the Z-axis vibration indicates that the gearbox may be experiencing stronger axial or vertical mechanical stress compared to the other directions. The sudden increase after 2029 may signify accelerated deterioration caused by imbalance, bearing wear, shaft misalignment, or lubrication problems. The recurring oscillations likely correspond to normal operational cycles, while the increasing instability indicates worsening machine condition. Overall, the graph suggests that the gearbox is moving toward a higher risk of mechanical failure and requires preventive maintenance and continuous monitoring.

Figure 3 presents the strong relationship between oil temperature and bearing temperature indicates that thermal conditions within the gearbox are closely interconnected. The near-perfect correlation among the vibration axes suggests that the vibrations may originate from the same mechanical source or system behavior. The negative relationship between oil pressure and vibration implies that reduced lubrication effectiveness may contribute to increased mechanical vibration and wear. Furthermore, the positive correlation between bearing temperature and vibration supports the idea that friction and mechanical stress simultaneously increase heat and vibration levels. Therefore, the heatmap confirms that lubrication condition, thermal behavior, and vibration are strongly linked indicators of gearbox health and performance.

Figure 4 indicates the higher median vibration and wider spread in the Z-axis suggest that this direction experiences the most severe mechanical stress or instability within the gearbox. The frequent outliers indicate intermittent abnormal conditions such as impacts, looseness, imbalance, sudden load changes, or bearing defects. Positive skewness in the vibration distributions implies that although the gearbox generally operates under normal vibration levels, occasional severe vibration events occur that may threaten machine reliability. The boxplot therefore reinforces the conclusion that the Z-axis is the most critical vibration direction and should receive greater attention during maintenance and condition monitoring activities.

C. Description of the Animated Graph Output

Figure 5 shows the separation between normal and anomalous data points indicates that the anomaly detection model can effectively distinguish irregular operating conditions from standard gearbox behavior. The anomalous points suggest unusual vibration responses that may be associated with early-stage mechanical faults, lubrication issues, bearing wear, or sensor irregularities. Since the anomalies occur within a relatively narrow temperature range, the results imply that abnormal gearbox conditions are not solely dependent on temperature increases but may also involve subtle changes in vibration characteristics. The visualization demonstrates the practical value of combining temperature and vibration features for predictive maintenance. Operators can use this information to identify developing faults before severe mechanical failure occurs, thereby reducing maintenance costs and improving wind turbine reliability.

Figure 6 presents the increasing vibration trend suggesting progressive mechanical degradation within the gearbox system. The sudden rise in vibration amplitudes around 2029 may indicate the onset of critical gearbox faults such as gear tooth wear, shaft imbalance, bearing damage, or misalignment. The consistently higher vibration levels in the Z-axis direction imply that vertical vibration components are more sensitive to structural or mechanical changes within the gearbox assembly.

The cyclical behavior observed throughout the monitoring period reflects normal operational loading patterns; however, the long-term increase in amplitude highlights the importance of continuous monitoring for early fault prediction. These results confirm that time-series vibration analytics can provide valuable insights into gearbox health and support condition-based maintenance scheduling. By identifying abnormal trends early, operators can perform targeted inspections and preventive maintenance actions before catastrophic failure occurs.

IV. CONCLUSION

This study successfully developed a vibration-based anomaly detection system for wind turbine gearbox condition monitoring using

engineering data analytics and machine learning techniques. The proposed system integrated automated data preprocessing, statistical analysis, anomaly detection, and visualization into a unified workflow capable of evaluating gearbox operational behavior efficiently. Results from the statistical summaries, correlation analysis, and visualization outputs revealed increasing vibration trends and operational instability, particularly along the Z-axis vibration component. The Isolation Forest algorithm effectively identified abnormal operating conditions associated with excessive vibration, thermal irregularities, and possible mechanical degradation. These findings demonstrate that combining vibration monitoring with machine learning analytics can improve predictive maintenance capability, reduce unexpected failures, and enhance wind turbine reliability.

The developed system also demonstrated strong scalability and adaptability for future industrial monitoring applications. Its modular Python-based architecture allows integration with larger renewable energy monitoring infrastructures, including SCADA systems, cloud-based analytics platforms, and real-time telemetry environments. Interactive dashboards and automated file generation improved engineering interpretability and workflow efficiency, enabling maintenance personnel to perform faster diagnostics and condition-based maintenance planning. Future enhancements may include real-time streaming analytics, advanced deep learning algorithms, and adaptive anomaly detection techniques to further improve fault prediction accuracy and support large-scale renewable energy system deployment.

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